

# Species Distribution Modeling for Eastern Oyster Habitat

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# 1 Introduction

The eastern oyster (*Crassostrea virginica*) are viewed as a foundational species as they are directly responsible for creating and providing numerous ecosystem goods and services such as food and habitat production, shoreline stabilization and protection, water quality improvement, nutrient cycling, and carbon sequestration (Geselbracht et al. 2024; Grabowski et al. 2012; Barnes et al. 2007). Additionally, they provide numerous economic benefits such as supporting oyster fisheries, recreational harvesting, tourism, along with the associated employment that goes along with those industries (Geselbracht et al. 2024; Peyre, Marshall, and Sable 2021). In one study, Grabowski et al. (2012) estimated the average annual value to range from \$10,325 to \$99,421 per hectare (2011 Dollars).

In recent decades, oyster reefs have experienced substantial declines, with only an estimated 15% remaining worldwide, and 50 to 80% being lost within the Gulf of (America) alone (Pollack et al. 2012; Peyre, Marshall, and Sable 2021). The decline is largely attributed to a combination of anthropogenic activities such as over harvesting, habitat alteration, sedimentation and water quality decline, disease and changing hydrology (Peyre, Marshall, and Sable 2021; Geselbracht et al. 2024). In response to these sharp declines, many restoration efforts have been undertaken with varying strategies and success rates; recently, planning documents released by the Deepwater Horizon Natural resource Damage Assessment Trustees called for large-scale development of reef networks to increase resilience (Geselbracht et al. 2024).

## 1.1 Oyster decline in the Pensacola Bay System

One such system to experience oyster decline is the Pensacola Bay System (PBS). The PBS is a large estuarine system located in Escambia and Santa Rosa counties in northwest Florida. The system is comprised of multiple waterbody's including Pensacola Bay, Escambia Bay, Blackwater Bay, East Bay, and Santa Rosa Sound; and is influenced heavily by freshwater input from the Escambia, Blackwater and Yellow Rivers (Christensen et al. 1997). Historically, the PBS supported extensive oyster reefs that provided critical habitat for a variety of species and substantial economic value to the surrounding communities (Christensen et al. 1997). However, over the past several decades, the extent of oyster habitat within the PBS has declined significantly due to a combination of factors including over harvesting, habitat degradation, and water quality issues. With some estimates indicating a decline of 15,000ha to just 91.5 ha of oyster habitat between the years of 1883 and 2021 (Geselbracht et al. 2024). This decline has prompted efforts to restore oyster reefs within the PBS (Pensacola and Perdido Bays Estuary Program n.d.; The Nature Conservancy 2021), with multiple studies aiming to identify the

most suitable areas within the estuary to focus restoration efforts (Geselbracht et al. 2024; Christensen et al. 1997; Pensacola and Perdido Bays Estuary Program n.d.).

With limited resources available for restoration, it is critical to identify areas where chances of oyster survival and long-term reef establishment are at their highest, and where poor site selection could ultimately lead to unsatisfactory results. As with any ecosystem, the environmental conditions within the PBS vary greatly in time and space, and such variations can have great impacts on oyster survival and establishment. Therefore, it is important to understand the environmental factors that drive oyster occurrence so that scientist and policy makers can make informed decisions regarding said restoration site selection (Pollack et al. 2012; Barnes et al. 2007; Geselbracht et al. 2024).

## **1.2 Environmental controls on Eastern oyster habitat**

Oyster recruitment, survival, and reef establishment are influenced by complex interactions between numerous physical, chemical, and biological environmental factors. As such, determining the most influential factors is a multivariate problem that requires careful consideration of each factor and its interactions with others (Peyre, Marshall, and Sable 2021; Pollack et al. 2012; Barnes et al. 2007; Christensen et al. 1997). To help narrow the focus, the most commonly cited environmental factors were examined in the literature and are summarized here.

Perhaps the most influential and widely reported environmental driver of oyster suitability is salinity. Representing an important environmental gradient within estuarine systems that influences physiology, recruitment, disease, predation, and survival. Other important characteristics often reported can be seen in **Table 1: Environmental drivers** and include: dissolved oxygen, temperature, depth, turbidity, and bottom type (substrate) (Peyre, Marshall, and Sable 2021; Pollack et al. 2012; Barnes et al. 2007; Geselbracht et al. 2024; Christensen et al. 1997).

Table 1: Common environmental drivers of Eastern Oyster habitat suitability.

| Factor                            | Description                                                                                                                                            | Sources                                                                                                                                         |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Salinity                          | Controls feeding, growth, reproduction, recruitment, disease, predation, and survival.                                                                 | (Barnes et al. 2007; Birch et al. 2021; Christensen et al. 1997; Geselbracht et al. 2024; Peyre, Marshall, and Sable 2021; Pollack et al. 2012) |
| Temperature                       | Regulates metabolism, filtration, growth, spawning, and larval development.                                                                            | (Barnes et al. 2007; Christensen et al. 1997; Geselbracht et al. 2024; Peyre, Marshall, and Sable 2021; Pollack et al. 2012)                    |
| Substrate type                    | Stable hard substrate supports larval attachment and reef persistence, whereas soft or rapidly accumulating sediments can bury or destabilize oysters. | (Barnes et al. 2007; Birch et al. 2021; Christensen et al. 1997; Geselbracht et al. 2024; Peyre, Marshall, and Sable 2021)                      |
| Dissolved oxygen                  | Adequate oxygen supports respiration, feeding, and growth.                                                                                             | (Christensen et al. 1997; Geselbracht et al. 2024; Peyre, Marshall, and Sable 2021; Pollack et al. 2012)                                        |
| Hydrology / water movement        | Water movement regulates salinity and transports larvae, oxygen, and food.                                                                             | (Barnes et al. 2007; Geselbracht et al. 2024; Peyre, Marshall, and Sable 2021)                                                                  |
| Depth / bathymetry                | Depth influences exposure, temperature, oxygen conditions, and restoration feasibility.                                                                | (Christensen et al. 1997; Geselbracht et al. 2024; Pollack et al. 2012)                                                                         |
| Turbidity / sedimentation         | High suspended-sediment loads can interfere with filtration and settlement. Sediment deposition can smother oyster reefs.                              | (Peyre, Marshall, and Sable 2021; Pollack et al. 2012)                                                                                          |
| Food availability / chlorophyll a | Chlorophyll a is often used as a proxy for food availability.                                                                                          | (Geselbracht et al. 2024; Peyre, Marshall, and Sable 2021)                                                                                      |

It should be recognized that not all environmental factors are adequately spatially distributed or represented within the available datasets, and therefore careful consideration should be given when selecting the final dataset used for modeling habitat suitability (Peyre, Marshall, and Sable 2021; Geselbracht et al. 2024; Christensen et al. 1997).

### **1.3 Environmentally Driven Habitat Modelling**

Quantifying the relationships between the distribution of species and the species-environment interactions have a long history in ecological research, having been described as far back as the late 17th early 18th century in which researchers observed biological patterns in terms of their relationships with geographical and/or environmental gradients (Rushton, Ormerod, and Kerby 2004; Elith and Leathwick 2009). Since then, the propagation of large-scale spatial datasets, the development and integration of GIS, increased access to statistical packages and computational power have allowed for much more development in modeling complex species-environment interactions (Johnson and Gillingham 2005).

#### **1.3.1 Habitat Suitability Indices (HSIs)**

HSIs are typically used in situations where quantitative data on species distributions is difficult to collect or non-existent, and they rely on expert-driven variable selection, meta-analysis, and classification methods that transform environmental characteristics into standardized estimates of habitat suitability, and have often been used as decision support tools in restoration frameworks due to their relatively low cost and flexibility (Barnes et al. 2007; Pollack et al. 2012; Geselbracht et al. 2024; Johnson and Gillingham 2005). Such models have already been used substantially in Eastern oyster restoration through out the Gulf Coast region in multiple different systems showing their applicability across spatial scales (Barnes et al. 2007; Pollack et al. 2012; Geselbracht et al. 2024; Christensen et al. 1997).

Within the PBS, HSIs have been used to develop GIS-based and estuary-specific HSMs that integrate biological, physical, chemical, and management-related variables in order to identify suitable restoration areas with the system. In the mid 1990's Christensen et al. (1997) developed a HSI for the eastern oyster in the PBS to demonstrate the model frameworks usefulness in illustrating species suitability in a system wide approach. Geselbracht et al. (2024) then followed and further expanded on the approach incorporating management factors into the model, adding a layer of regulatory influence to the model output versus the purely environmental analysis of the prior. Additionally, the Pensacola and Perdido Bay Estuary Program recently built on-top of Geselbracht et al. (2024) by incorporating some of their own data and refining some of the weighting (Pensacola and Perdido Bays Estuary Program n.d.).

These previous studies alone highlight the strengths and weakness of the HSI approaches. Some of the strengths include interoperability, adaptability, and the integration of ecological knowledge, but the results entirely rely on expert decisions regarding variable selection, suitability functions,

weighting and scoring, and how to aggregate the results. All of these present opportunity for differing outputs even if the same inputs were used by different groups (Geselbracht et al. 2024; Johnson and Gillingham 2005).

### **1.3.2 Species Distribution Modelling**

Another category of habitat modeling commonly used within the field and be broadly referred to as species distribution models (SDMs) can be broadly defined as numerical tools that combine observations of species occurrence or abundance with environmental estimates (Elith and Leathwick 2009). When combined with Geographic Information Systems (GIS), these predictive models have the ability to visualize the distribution and infer the suitability of species habitats. However, as with any statistical modeling techniques, different approaches can generate different predictions, and with differing modeling methods and environmental inputs able to influence the predictions, careful effort should be given when choosing which model to employ (Johnson and Gillingham 2005).

Ultimately, it is now well understood that empirically based SDMs can provide a reliable and alternative method of suitability modeling based off of environmental relationships observed directly using known occurrence data (Phillips, Anderson, and Schapire 2006; Rushton, Ormerod, and Kerby 2004; Elith and Leathwick 2009; Johnson and Gillingham 2005). One such technique that has come to be very prominent in the realm of SDM and ecology in general is known as the MaxEnt approach (Phillips, Anderson, and Schapire 2006; Merow, Smith, and Silander 2013).

### **1.3.3 Maximum Entropy species distribution modeling**

1. Introduce MaxEnt as a presence-background species distribution modeling framework.
  - Particularly useful where reliable absence observations are unavailable.
  - Relates environmental conditions at known occurrence locations to environmental conditions available throughout the modeling region.
  - Citations: (Phillips, Anderson, and Schapire 2006; Elith et al. 2011)
2. Explain why MaxEnt is appropriate for Eastern oyster occurrence data in PBS.
  - FWC oyster-bed data provide known oyster occurrences without systematically sampled absence locations.
  - Presence-background modeling therefore fits the available data structure.
  - Citations: (Phillips, Anderson, and Schapire 2006; FWC-FWRI 2026)
3. Explain important considerations when implementing MaxEnt.
  - Background extent and sampling.
  - Spatial sampling bias.

- Predictor selection.
  - Regularization.
  - Feature classes and model complexity.
  - Citations: (Phillips et al. 2009; Merow, Smith, and Silander 2013; Elith et al. 2011)
4. Explain the need for model tuning and spatial validation.
- MaxEnt should not be treated as an automated black-box model.
  - Model configuration should be evaluated systematically.
  - Spatially independent validation should be used where occurrence data exhibit spatial structure.
  - Citations: (Kass et al. 2021; Roberts et al. 2017; Valavi et al. 2019)

## **1.4 Research rationale, objectives, and hypotheses**

1. Establish the primary research problem.
  - An estuary-specific HSI already exists for Eastern oyster restoration within PBS.
  - This study will independently develop an empirically fitted MaxEnt habitat-suitability model using observed oyster occurrences and environmental conditions.
  - Citations: (Geselbracht et al. 2024; Phillips, Anderson, and Schapire 2006)
2. Primary objective.
  - Develop a spatially explicit MaxEnt SDM representing relative Eastern oyster habitat suitability throughout PBS.
3. Secondary objective.
  - Quantitatively compare the MaxEnt suitability surface with the existing Pensacola Bay HSI.
4. Supporting objectives.
  - Identify environmental predictors associated with observed oyster distribution.
  - Evaluate MaxEnt predictive performance using spatially separated validation data.
  - Identify geographic areas where MaxEnt and the HSI agree or disagree.
  - Evaluate whether MaxEnt provides a reproducible empirical complement to HSI-based restoration planning.
  - Citations: (Johnson and Gillingham 2005; Geselbracht et al. 2024; Merow, Smith, and Silander 2013)
5. Research hypothesis.
  - The MaxEnt model is expected to identify patterns of habitat suitability consistent with established Eastern oyster habitat requirements.

- Measurable agreement is expected between MaxEnt and the existing HSI, with spatial disagreement arising from differences between empirically fitted environmental relationships and predefined HSI suitability relationships.
- Citations: (Johnson and Gillingham 2005; Geselbracht et al. 2024)

## 2 Methods

### 2.1 Data Processing and Analysis

#### 2.1.1 Software and Packages

All data processing, analysis, and modeling will be conducted using free open source software and packages to maximize accessibility and reproducibility. The majority of data manipulation and processing will be performed in the R programming language using the tidyverse suite of packages (Wickham et al. 2019; R Core Team 2024). Any data operations that need to be “spatially aware” will be performed using the Simple Features package (Pebesma 2018).

Analysis will be scripted and ran in Positron, an open-source integrated development environment developed by Posit that supports R, Python, scripts, notebooks, integrated consoles, along with interactive plots and data viewers.

#### 2.1.2 Study Area

The study area for this project is the Pensacola Bay System which includes portions of Pensacola Bay, Escambia Bay, East Bay and Blackwater Bay. Boundaries of this study area were achieved by accessing Florida Department of Protections (FDEP) Waterbod IDs feature layer hosted on ArcGIS Online (FDEP 2005). ArcGIS allows direct access to hosted feature services via several formats. To avoid any potential compatibility issues, the GeoJSON format was chosen as this format is non-proprietary. Once retrieved, the dataset was filtered to include only relevant WBIDs (WBID = 548) and then combined to create a singular study area polygon. Creating the study area polygon is an important first step as it will be needed to extract relevant environmental data from much larger datasets in the coming data gathering.

#### 2.1.3 Water Quality Portal Data

Water quality data is widely available for the entire United States via the EPA’s Water Quality Portal (WQP) which allows for a wide range of data querying for any data as it relates to Water Quality Monitoring programs (USGS (2026)). For this analysis, site data for Escambia and Santa Rosa County was queried using the WQP API then the study area polygon was used to extract only the relevant sites:

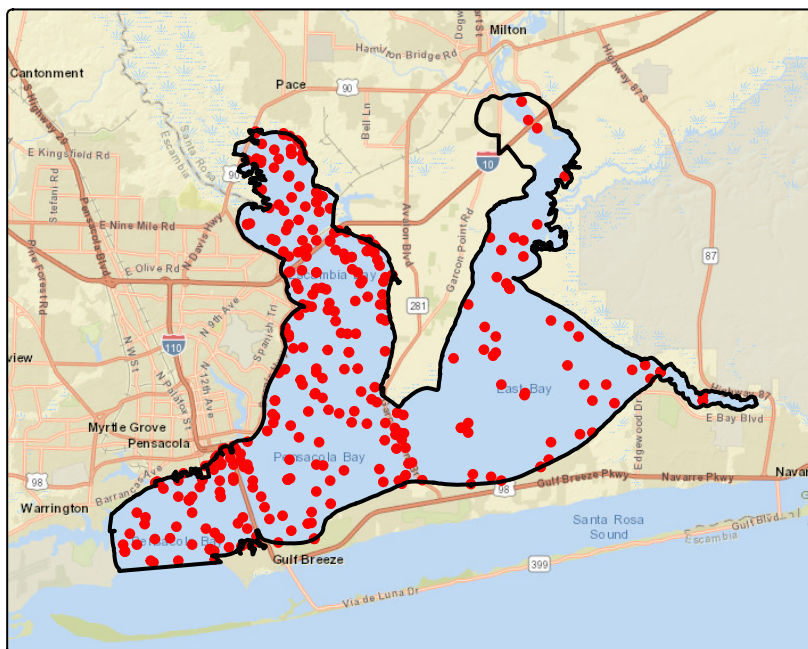


Figure 1: Pensacola Bay System with monitoring sites

#### 2.1.4 Water Quality Data

Water quality data was selected based off of availability and relevance within literature. Each parameter was filtered to include just those measurements

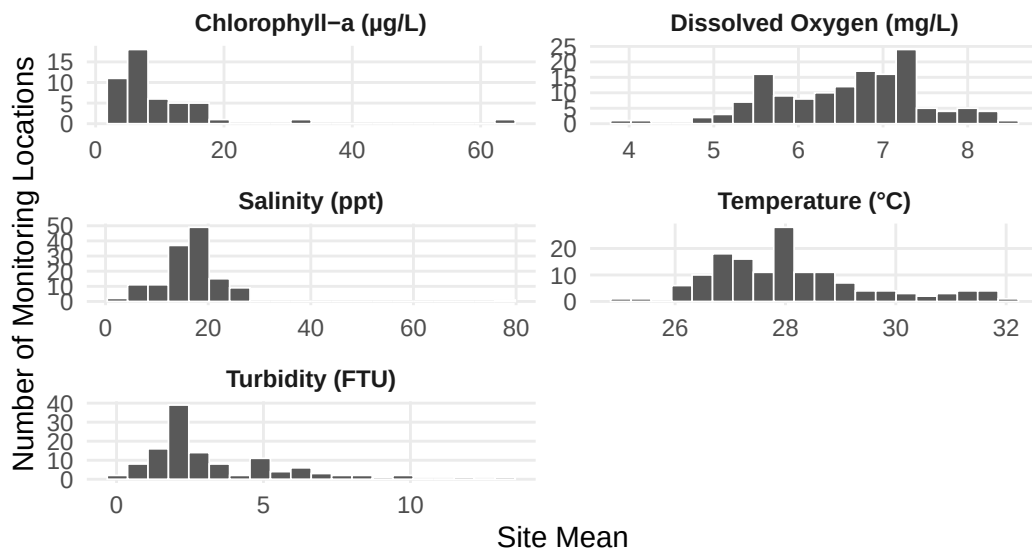
Table 2: Preliminary Environmental Variables Considered for the Eastern Oyster SDM

| Variable                 | Obtained?<br>(Y/N) | Data<br>Type | Total Records                                         | Data Source                                      |
|--------------------------|--------------------|--------------|-------------------------------------------------------|--------------------------------------------------|
| Salinity                 | Y                  | Points       | 6994                                                  | National Water Quality Monitoring Council (2026) |
| Water temperature        | Y                  | Points       | 7161                                                  | USGS (2026)                                      |
| Substrate / bottom type  | Y                  | Raster       | NA                                                    | (ppbep?)                                         |
| Water depth / bathymetry | N                  | Raster       | National Centers for Environmental Information (2018) | NA                                               |

| Variable             | Obtained?<br>(Y/N) | Data<br>Type | Total Records      | Data Source |
|----------------------|--------------------|--------------|--------------------|-------------|
| Dissolved oxygen     | Y                  | Points       | 6261               | USGS (2026) |
| Chlorophyll-a        | Y                  | Points       | 1847               | USGS (2026) |
| Oyster bed locations | NA                 | NA           | FWC-FWRI<br>(2026) | NA          |

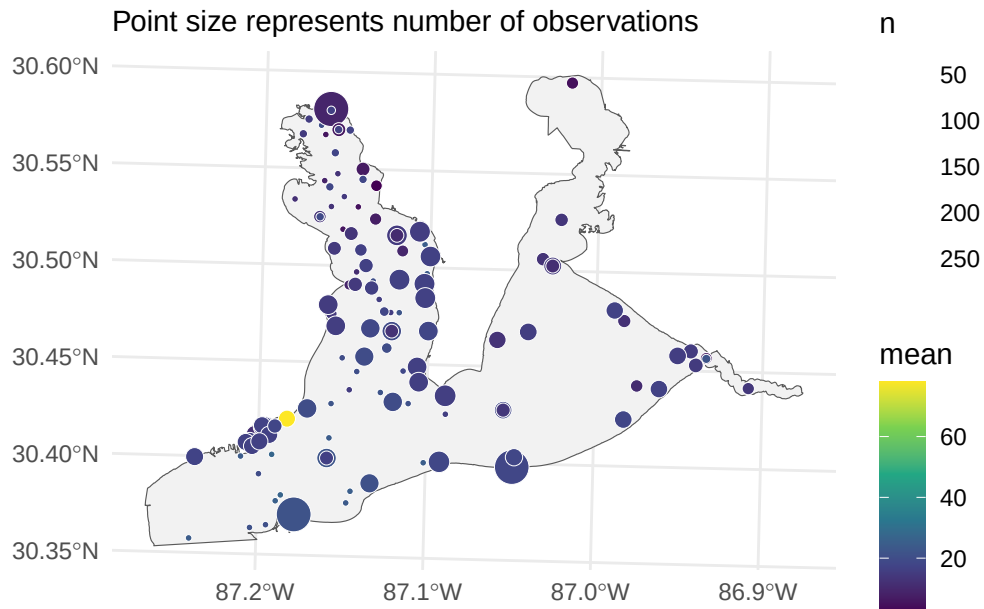
## Distribution of Warm-Season Water Quality Conditions

Site-level means for monitoring locations in the Pensacola Bay System



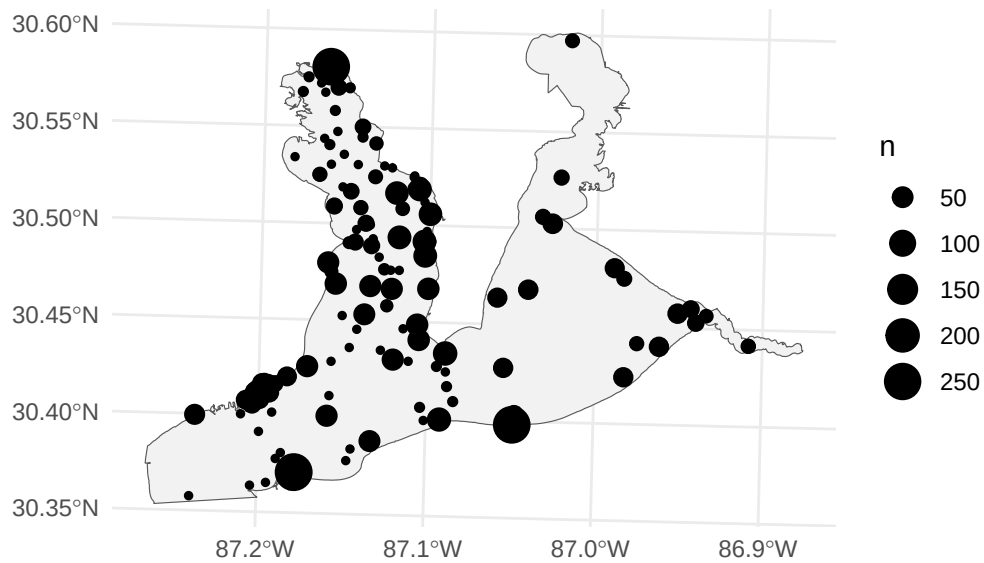
### Warm-Season Salinity Monitoring Locations

Point size represents number of observations



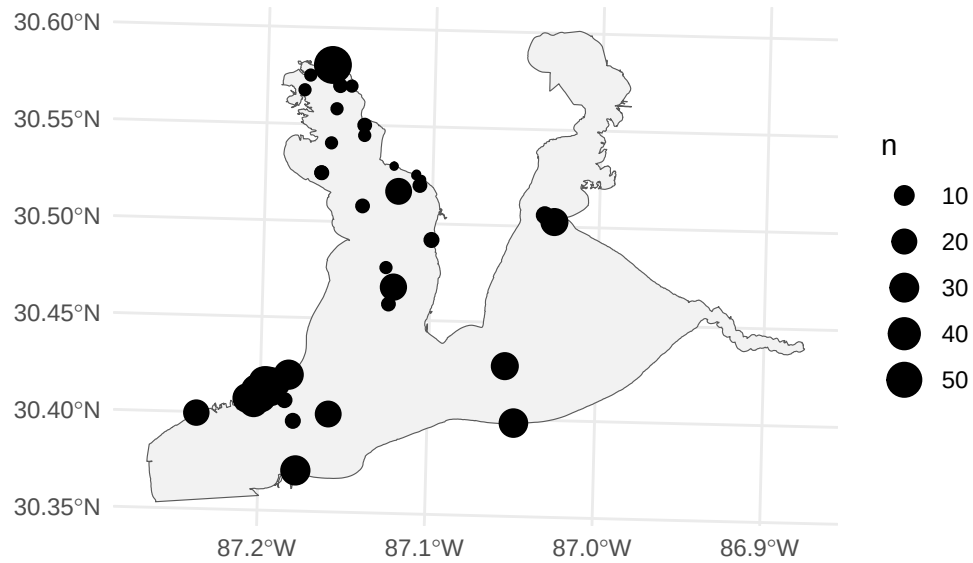
### Warm-Season Temperature Monitoring Locations

Point size represents number of observations



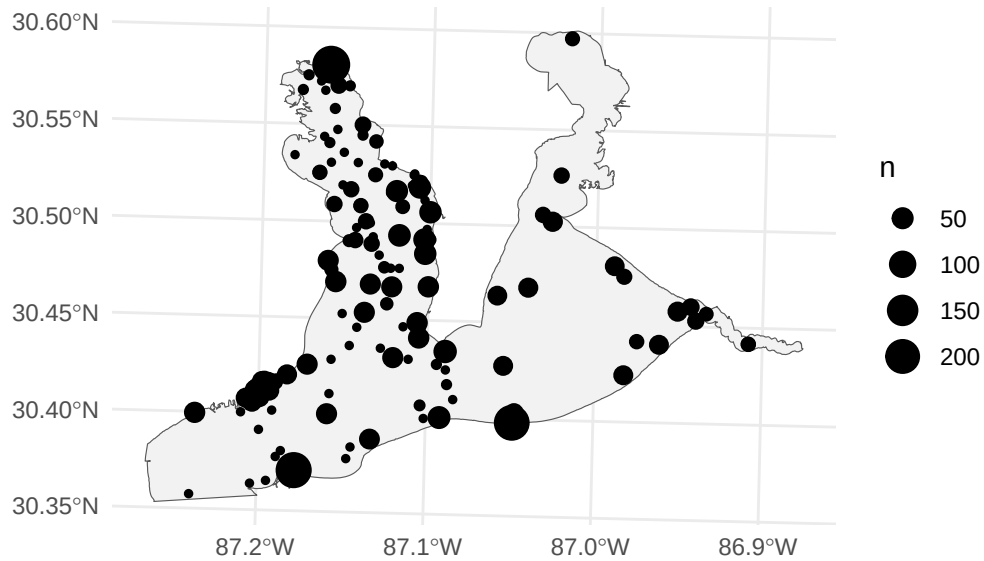
### Warm-Season Chlorophyll-a Monitoring Locations

Point size represents number of observations



### Warm-Season Dissolved Oxygen Monitoring Locations

Point size represents number of observations



## Warm-Season Turbidity Monitoring Locations

Point size represents number of observations

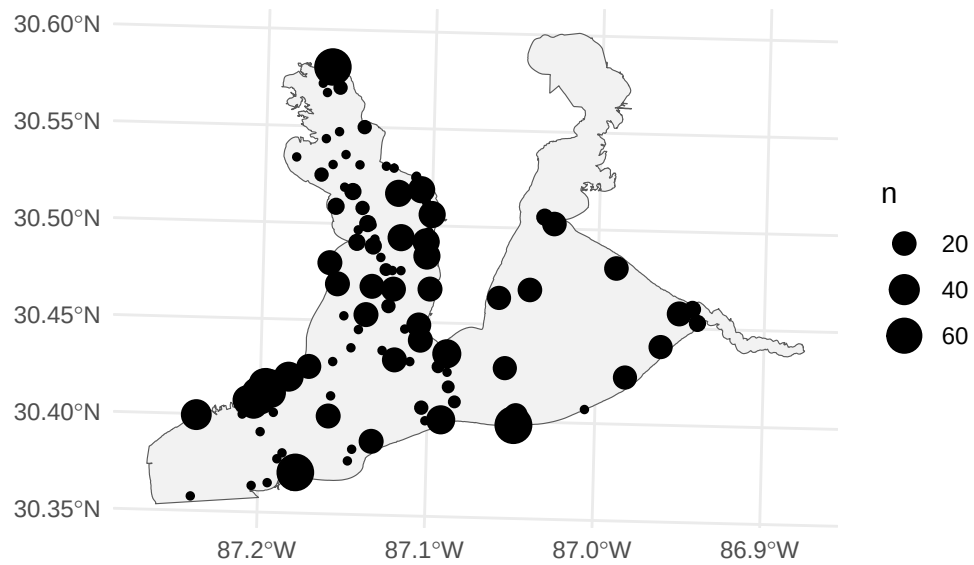


Table 3: Summary Statistics for Warm-Season Water Quality Data

| Parameter                  | Sta-<br>tions | Observa-<br>tions | Mean  | SD   | Min   | Q1    | Me-<br>dian | Q3    | Max   | First<br>Sample | Last<br>Sample |
|----------------------------|---------------|-------------------|-------|------|-------|-------|-------------|-------|-------|-----------------|----------------|
| Chlorophyll-a<br>(µg/L)    | 48            | 754               | 9.74  | 9.70 | 3.39  | 5.14  | 6.21        | 11.79 | 63.80 | 2005-05-03      | 2024-10-17     |
| Dissolved Oxygen<br>(mg/L) | 145           | 3072              | 6.59  | 0.87 | 3.90  | 5.82  | 6.75        | 7.21  | 8.48  | 2005-05-03      | 2024-10-17     |
| Salinity (ppt)             | 135           | 3401              | 16.55 | 7.35 | 2.31  | 14.02 | 16.54       | 18.91 | 77.85 | 2005-05-03      | 2024-10-17     |
| Temperature (°C)           | 145           | 3442              | 28.11 | 1.43 | 25.01 | 27.05 | 27.92       | 28.74 | 32.04 | 2005-05-03      | 2024-10-17     |
| Turbidity (FTU)            | 122           | 1563              | 3.33  | 2.41 | 0.03  | 1.85  | 2.35        | 4.65  | 13.22 | 2005-05-10      | 2024-10-17     |

## 2.2 Spatial interpolation

Points are interpolated into prediction rasters using inverse path distance weighting (IPDW) (package idpw). Blah blah blah blah

## 3 Results

In progress...

## 4 Discussion

In progress...

## 5 Conclusion

In progress...

## 6 References

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